

AUDIOBOOK OS

Project Requirements, Functional Specification & Technical Specification

A build-ready baseline for a working audiobook production tool

Build baseline

Ship a reliable publisher tool first: ingest, script engineering, provider-independent synthesis, text-first punch-in editing, automated QC, mastering, approval, and export. Add configurable workflows, distribution, plugins, marketplace, and multi-tenant SaaS in later waves.

Prepared for	Mangu Publishers
Product	Audiobook OS
Document version	1.0
Status	Implementation baseline
Date	July 23, 2026

Document Control

Field	Baseline
Purpose	Define the product scope, functional behavior, technical architecture, data contracts, acceptance criteria, and phased delivery plan required to build a working audiobook production tool.
Primary audience	Product, engineering, audiobook production, QA, security, legal/rights, and implementation partners.
Source inputs	Audiobook OS · Platform Blueprint v2.0 and ACX NATION.docx research, workflow, full-stack review, and question bank.
Decision posture	Use licensed / hosted speech providers for the first working releases; build the orchestration, editor, QC, mastering, rights, and production record as the defensible product.
Change control	Changes to MUST requirements, release gates, export profiles, provider contracts, or rights controls require documented approval and a version increment.

How to read the specification

Requirements and source-backed product principles are stated as the baseline. Concrete implementation choices are labeled “Proposed implementation.” Items that depend on changing retailer policy, official audio specifications, provider capability, or legal review are labeled “Release gate” and must be revalidated before production use.

Contents

Section	Covers
Executive Summary	Build decision, scope, and definition of a working tool
1. Product & Project Requirements	Goals, users, scope, success measures, constraints
2. End-to-End Operating Workflow	Production stages, gates, and persistent records
3. Functional Specification	Build requirements by product capability
4. User Experience Specification	Workspace, screens, and core interactions
5. Technical Architecture	Components, service boundaries, jobs, storage, events
6. Domain & Data Model	Core entities, versioning, and lifecycle rules
7. API & Event Contracts	Resource endpoints and domain events
8. Provider & Extension Contracts	Speech-provider, AI-agent, and distribution interfaces
9. Audio Pipeline & Export Profiles	Processing chain, validation, and ACX research profile
10. Security, Rights & Governance	Access control, consent, disclosure, audit, budgets
11. Non-Functional Requirements	Reliability, performance, accessibility, scalability
12. Test & Acceptance Strategy	Automated, audio, contract, and user acceptance gates
13. Delivery Roadmap	Pilot, working release, production hardening, platform waves
14. Risks & Open Decisions	Known uncertainties and required decisions
15. Definition of Done	Working tool and long-term platform completion tests
Appendices	Traceability, terminology, and approval record

Executive Summary

The two source documents point to the same conclusion: the opportunity is not a simple “upload text and download MP3” utility. The durable product is a publisher-grade production system that preserves every manuscript revision, voice decision, synthesis attempt, edit, QC finding, approval, export, and distribution event as one traceable audiobook production record.

Strategic build decision

Do not make proprietary speech-model research the critical path for the first working tool. License at least one high-quality long-form provider, keep AWS Polly as a cost-effective baseline adapter, and invest engineering in the text-first editor, sentence-level punch-in regeneration, automated transcript-diff QC, mastering, rights/disclosure controls, and provider-independent production workflow.

What the first working release must do

- Create a project with target retailers, narration mode, budget, rights, consent, and disclosure requirements.
- Ingest a long-form manuscript, preserve the immutable source, detect chapters and structure, and let a human correct the production script.
- Build pronunciation and character/voice guides, preview voices, and assign provider-independent voice profiles.
- Generate audio asynchronously at segment, chapter, or project scope with progress, cancellation, retries, provenance, and cost estimation.
- Provide a text-aligned editing workspace where a producer can select one sentence, regenerate it in context, compare takes, and commit a non-destructive edit.
- Run automated technical and text-to-audio QC, route findings through human review, and prevent approval while blocking issues remain.
- Master and package approved audio through versioned export profiles, produce a validation report, and retain the evidence needed for retailer submission.

What is deliberately deferred

- Training a proprietary long-form voice model or the separate 100-knob physical-modeling voice instrument.
- Soundscaping as a default production layer; it remains an optional future capability pending demand, intelligibility, licensing, and workflow research.
- Marketplace, third-party plugin commercialization, public GraphQL, multi-language SDKs, tenant billing, and full multi-tenant SaaS operations.
- Automated direct distribution to every retailer where official or stable submission APIs are not available; R1 produces a compliant package and evidence bundle.

Release ladder

Release	Outcome
R0 • Evidence pilot	3–5 backlist titles using licensed providers. Measure cost, hours per finished hour, QC defect rate, punch-in continuity, listener response, and retailer acceptance.
R1 • Working publisher tool	Single-organization production workspace covering ingest through approved export; persistent state, jobs, editing, QC, mastering, rights, and audit.
R2 • Production platform	Provider routing/failover, configurable workflows/rules, distribution adapters, analytics, production timeline, hardened security and operations.

Release	Outcome
R3 · Audiobook OS	Plugins, marketplace, public API/SDKs, tenant isolation, quotas, billing, and ecosystem operations.

1. Product & Project Requirements

This section converts the platform vision and ACX NATION research into the bounded product the team should actually build. It separates the first working release from capabilities that belong to later platform waves.

1.1 Product objective

Product objective

Transform a final manuscript into a reviewable, rights-cleared, technically validated, retailer-ready audiobook while minimizing repeated work, making every decision auditable, and keeping speech providers replaceable.

1.2 Primary users and roles

Role	Responsibilities
Producer	Creates projects, chooses workflow and targets, controls budget, assigns work, approves final master.
Script editor	Corrects structure, removes visual-only references, prepares narration script, speakers, pronunciation, and direction.
Voice director / casting lead	Auditions voices, assigns narrator/characters, manages voice bible and performance direction.
Audio editor / engineer	Reviews generated or recorded audio, performs punch-ins, assembly, mastering, and technical correction.
QC reviewer / proofer	Checks transcript alignment, pronunciation, consistency, completeness, pacing, and technical compliance.
Legal / rights reviewer	Confirms manuscript rights, narrator/voice consent, allowed uses, disclosure, and publishing approval.
Administrator	Manages users, providers, secrets, defaults, budgets, retention, and audit access.
External reviewer (R2)	Read-only or comment access to assigned chapters, issues, or production timeline.

1.3 In scope for R1

- One organization with project-level role-based access; data structures retain organization scoping so multi-tenancy can be added later.
- Text/DOCX/EPUB and text-based PDF ingestion; scanned-PDF OCR is R2 unless required by the pilot.
- Synthetic narration and hybrid human + AI pickup workflows; human recording intake remains supported as an audio asset source.
- At least two speech-provider adapters: AWS Polly baseline plus one premium long-form provider selected in the pilot.
- Text-first production editor, segment regeneration, take comparison, non-destructive edit decisions, chapter assembly, automated QC, human review, mastering, export, and evidence bundle.

- Versioned retailer export profiles and disclosure templates, with policy/spec verification as a release gate rather than hardcoded assumptions.

1.4 Out of scope for R1

- Training or hosting a proprietary foundational TTS model.
- Unconsented voice cloning, posthumous voice use, or any voice use lacking a complete rights record.
- Full-cast automated dramatic direction without human review.
- Generative ambience/SFX, adaptive soundscape mixing, or music licensing workflows.
- Marketplace billing, public partner certification, customer self-service tenancy, and white-label domains.

1.5 Success measures

Measure	Target outcome
Pilot completion	Produce 3–5 backlist titles end to end and record cost, elapsed time, human hours, defects, corrections, and acceptance outcome.
Completeness	A 90k–100k word manuscript can be imported, segmented, generated, and exported with no missing chapter or paragraph.
Correction speed	A reviewer can identify and replace one sentence without regenerating the whole chapter; the chosen take is retained with provenance.
Quality gate	No export can be approved with unresolved blocking QC, incomplete rights, or missing channel disclosure.
Technical evidence	Every export includes a machine-readable and human-readable validation report for the selected profile.
Provider independence	The same project can render a preview through either configured provider without changing core domain code.
Traceability	A producer can reconstruct who changed what, which provider/model generated each asset, what it cost, and why the final take was selected.

1.6 Design tenets

- Configuration over code: export profiles, disclosure templates, provider priority, budgets, and production rules are data.
- Immutable source, editable production script: automation never overwrites the uploaded manuscript.
- Provider independence: providers implement stable contracts and declare capabilities.
- Asynchronous by default: long-running work is queued, persistent, retryable, cancellable, and observable.
- Non-destructive production: edits select and assemble versioned assets instead of destructively rewriting audio.
- Governed by default: cost, rights, approval, disclosure, and tenant/organization boundaries are enforced in the backend.
- Everything meaningful emits an event: automation, audit, analytics, webhooks, and the production timeline share one source.

2. End-to-End Operating Workflow

The ACX NATION production model treats an audiobook as a media product, not a file conversion. R1 implements the full production record but keeps the workflow fixed enough to ship. R2 re-expresses the stages as configurable workflow definitions and rules.

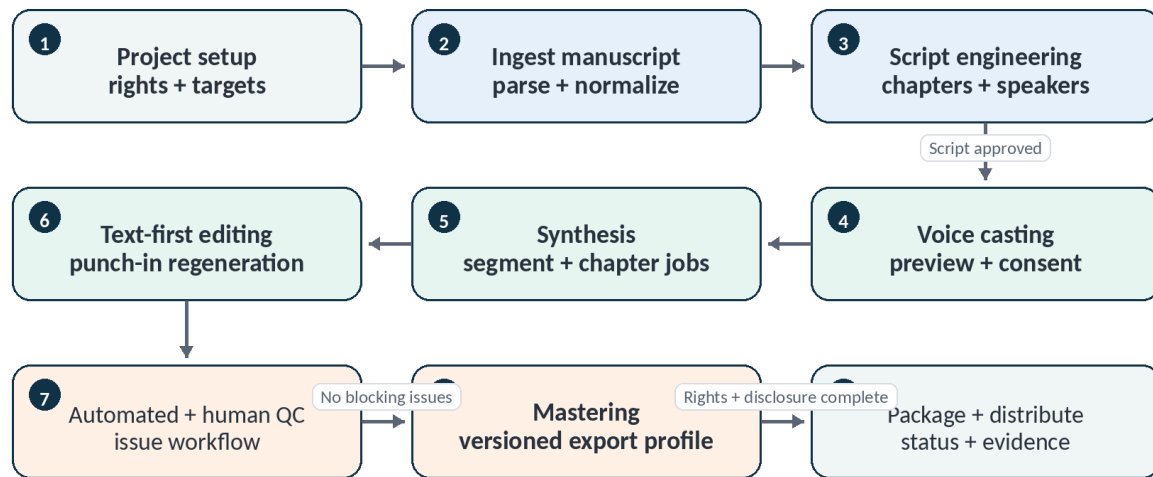


Figure 1. Canonical production flow for the working release. Every stage persists its inputs, outputs, decisions, and evidence.

Stage	Purpose	Persistent output	Exit gate
1. Project setup	Commercial and legal context	Targets, budget, narration mode, rights record, disclosure plan	Required fields complete
2. Ingest	Immutable source and parsed structure	Manuscript revision, metadata, chapters, warnings	Structure accepted by editor
3. Script engineering	Production-ready narration script	Cleaned script, speakers, pronunciation, direction	Script approved
4. Voice casting	Provider-independent voice plan	Previews, assignments, voice bible, consent evidence	Voice and rights approved
5. Synthesis	Versioned generated takes	Jobs, attempts, word timings, cost, audio assets	Required segments complete
6. Text-first editing	Selected final takes and edit decisions	EDL, crossfades, corrected segments, human pickups	Chapter edit locked
7. QC	Automated and human findings	QC runs, issues, resolutions, reviewer sign-off	No blocking issues
8. Mastering	Profile-compliant master	Master WAV, chapter files, metrics, validation report	Profile pass
9. Package / distribute	Retail submission asset	Metadata, cover, rights, disclosure, export version, status	Publishing approval

Non-negotiable workflow rule

A stage may recommend or prepare the next action, but a paid operation, state transition, approval, or distribution action must pass the backend gate for budget, rights, disclosure, and unresolved blocking issues.

3. Functional Specification

Requirement IDs are stable references for product backlog, API stories, test cases, and acceptance. “MUST” requirements define the working release unless the Release column says R0/R2/R3.

3.1 Project setup, rights, and governance

ID	Requirement	Priority	Release	Acceptance test
FR-PRJ-001	Create and edit a project with title, author, language, expected word count/runtime, target channels, planned release, narration mode, and production owner.	MUST	R1	A saved project can be reopened with all fields and a unique project ID.
FR-PRJ-002	Maintain project status and stage history from draft through approved export; transitions are validated server-side.	MUST	R1	Invalid transitions return a typed error and do not alter state.
FR-PRJ-003	Capture manuscript rights, audio rights, territories, term, source evidence, and publishing approval.	MUST	R1	Synthesis/export is blocked when required rights fields or evidence are missing.
FR-PRJ-004	Capture narration mode as human, hybrid, or synthetic and maintain retailer-specific disclosure copy/status.	MUST	R1	An export cannot be approved without required disclosure for every selected channel.
FR-PRJ-005	Attach narrator/voice consent evidence, permitted uses, compensation model, credit, revocation terms, and restrictions for any cloned or replica voice.	MUST	R1	A cloned/replica voice cannot be assigned until its rights status is approved.
FR-PRJ-006	Estimate provider and processing cost before a project/chapter run and require approval above configured thresholds.	MUST	R1	User sees estimate, reusable segments, and approval requirement before commit.
FR-PRJ-007	Support role-based permissions for producer, editor, reviewer, legal, and administrator.	MUST	R1	Each protected command is allowed/denied by backend policy and audited.
FR-PRJ-008	Allow organization/project templates for default target channels, export profile, provider priority, and disclosure language.	SHOULD	R2	A new project inherits a versioned template and records any overrides.

3.2 Manuscript ingestion and script engineering

ID	Requirement	Priority	Release	Acceptance test
FR-CNT-001	Upload TXT, DOCX, EPUB, and text-based PDF files up to the configured limit; validate extension, MIME type, size, and parseability.	MUST	R1	Valid sample files import; invalid/oversize files are rejected with actionable errors.
FR-CNT-002	Store the original manuscript as an immutable, checksummed revision and create a separate editable production-script revision.	MUST	R1	Editing the production script never changes the original object or hash.
FR-CNT-003	Extract title/author/language where available and detect chapters, headings, front/back matter, paragraphs, and scene breaks.	MUST	R1	A 40-chapter test manuscript produces ordered chapters with source offsets and warnings.

ID	Requirement	Priority	Release	Acceptance test
FR-CNT-004	Let an editor split, merge, reorder, rename, include, or exclude chapters/sections before approval.	MUST	R1	Corrections persist as a new script revision and preserve source traceability.
FR-CNT-005	Flag visual-only references, tables, footnotes, URLs, page references, and other content that needs narration treatment.	MUST	R1	Flagged items appear in a review queue and require disposition.
FR-CNT-006	Identify narration, dialogue, quotes, poetry, credits, and candidate speakers, with human correction.	MUST	R1	Every production segment has a type and speaker assignment or an explicit unresolved flag.
FR-CNT-007	Maintain a project pronunciation dictionary with spelling, phonetic form, provider lexicon/SSML representation, notes, and approval.	MUST	R1	Approved entries are applied consistently to previews and final synthesis.
FR-CNT-008	Attach sentence/scene-level performance direction such as pacing, tone, mood, emphasis, pause, and emotional beat.	SHOULD	R1	Direction is stored separately from source text and included only when the provider supports it.
FR-CNT-009	Generate completeness checks comparing source, production script, chapters, and segments.	MUST	R1	The system identifies omitted or duplicated source ranges before synthesis approval.
FR-CNT-010	Support OCR ingestion for scanned PDFs with confidence and mandatory human verification.	COULD	R2	Low-confidence OCR content cannot advance without explicit review.

3.3 Voice casting, pronunciation, and direction

ID	Requirement	Priority	Release	Acceptance test
FR-VOI-001	List configured speech providers, voices, languages, styles, and declared capabilities such as SSML, word timestamps, long-form, cloning, and streaming.	MUST	R1	The UI only offers voices eligible for the project language and required capability set.
FR-VOI-002	Generate a short preview from user-selected text and settings without starting a full production job.	MUST	R1	Preview returns playable audio, provider/model provenance, and estimated unit cost.
FR-VOI-003	Assign a voice profile to narrator and each approved speaker/character; maintain a reusable narrator/character voice bible.	MUST	R1	Assignments persist across chapters and can be copied to a series project.
FR-VOI-004	Support rate, pitch, pause, style prompt, emotion/direction, lexicon, and SSML where the provider declares support.	MUST	R1	Unsupported controls are disabled or translated by the adapter with a visible warning.
FR-VOI-005	Require rights status and consent evidence before a cloned/replica/custom voice can be previewed or synthesized.	MUST	R1	Backend rejects the request when rights status is not approved.
FR-VOI-006	Allow imported human narration or pickup audio to be assigned to a segment in the same edit and QC workflow.	MUST	R1	Human and synthetic assets can be compared and selected without changing downstream contracts.
FR-VOI-007	Maintain primary and backup provider/voice mappings per role.	SHOULD	R2	A failed preferred provider can be replaced without editing the manuscript or speaker model.
FR-VOI-008	Preserve provider/model/version/voice/settings provenance for every preview and production take.	MUST	R1	An asset detail view reconstructs the exact generation request.

3.4 Synthesis, rendering, and job control

ID	Requirement	Priority	Release	Acceptance test
FR-SYN-001	Create synthesis jobs at segment, chapter, selected-chapter, or full-project scope.	MUST	R1	A user can generate one sentence or all approved chapters using the same job model.
FR-SYN-002	Queue long-running work outside the web process and persist job state, progress, errors, attempts, timestamps, and output references.	MUST	R1	A worker/server restart does not lose job state; interrupted jobs can resume or retry.
FR-SYN-003	Support cancel, retry, resume, and retry-from-failed-segment commands with idempotency protection.	MUST	R1	Repeated commands do not duplicate billable synthesis or assets.
FR-SYN-004	Use content-addressed reuse keyed by normalized text, voice, provider/model, pronunciation, direction, and relevant context.	MUST	R1	Unchanged approved segments are reused and reported in the cost estimate.
FR-SYN-005	Split long content at safe sentence/paragraph boundaries while preserving chapter order, source offsets, and context.	MUST	R1	No request exceeds provider limits and no split occurs mid-word or loses text.
FR-SYN-006	Capture speech/word timestamps or speech marks when available and normalize them to the platform timing model.	MUST	R1	The editor can highlight text during playback for providers that support timing.
FR-SYN-007	Store every provider attempt, response metadata, error class, billed units, latency, and resulting asset.	MUST	R1	Cost and failure reports reconcile to attempts and assets.
FR-SYN-008	Show estimate-before-commit: characters, expected duration, provider cost, reusable vs new segments, and completion range.	MUST	R1	Production cannot start until the user confirms or an approved automation rule does so.
FR-SYN-009	Select and fail over across eligible providers based on policy, language, capability, cost, quality, latency, and availability.	SHOULD	R2	A simulated provider outage routes new attempts to the next eligible provider and records the decision.
FR-SYN-010	Support parallelism within configured provider and organization quotas.	SHOULD	R2	Throughput increases without exceeding rate limits or reordering outputs.

3.5 Text-first editing and punch-in regeneration

ID	Requirement	Priority	Release	Acceptance test
FR-EDT-001	Display chapter script, sentence boundaries, speaker, waveform, playback cursor, timings, selected take, and QC markers in one workspace.	MUST	R1	Playback highlights the aligned text and issue markers seek to the correct time range.
FR-EDT-002	Allow sentence/segment selection and one-click preview or regeneration using surrounding context and current voice settings.	MUST	R1	Only the selected segment is regenerated; adjacent approved assets remain unchanged.
FR-EDT-003	Let a producer compare multiple takes, A/B play them in context, annotate them, and select the committed take.	MUST	R1	Selection creates a new edit decision; prior takes remain available.
FR-EDT-004	Represent chapter assembly as a non-destructive edit decision list over versioned assets, with trim points, gaps, fades, and ordering.	MUST	R1	Rebuilding a chapter from the EDL produces a deterministic result.

ID	Requirement	Priority	Release	Acceptance test
FR-EDT-005	Apply configurable crossfades and continuity checks at replacement boundaries.	MUST	R1	Punch-in joins are checked for abrupt level, timing, spectral, or silence discontinuity.
FR-EDT-006	Support undo/redo and revision history for script, assignment, and edit-decision changes.	MUST	R1	A user can restore a prior approved chapter revision without losing subsequent audit history.
FR-EDT-007	Provide keyboard-driven next-issue, play/pause, loop, mark, accept, and regenerate actions for high-throughput review.	SHOULD	R1	A reviewer can complete the defined proofing flow without using a mouse.
FR-EDT-008	Import and align a human pickup/recording to replace a generated segment.	SHOULD	R1	Imported audio receives the same QC, take comparison, and provenance treatment.

3.6 Audio engineering and mastering

ID	Requirement	Priority	Release	Acceptance test
FR-AUD-001	Normalize provider outputs into a lossless working format before assembly; avoid repeated lossy MP3 encode/decode cycles.	MUST	R1	A chapter is assembled from PCM/WAV intermediates and encoded once per final export.
FR-AUD-002	Provide configurable cleanup and mastering stages for trim, silence, fades, DC offset, resampling, noise, EQ, de-essing, dynamics, loudness, and limiting.	MUST	R1	Each enabled stage records parameters and before/after metrics.
FR-AUD-003	Control head/tail room tone, sentence/chapter gaps, and credit packaging by export profile.	MUST	R1	Generated chapter files meet the profile's gap and room-tone rules.
FR-AUD-004	Measure RMS/loudness, true peak, sample peak, noise floor, clipping, silence ratio, duration, sample rate, channels, codec, and bitrate.	MUST	R1	Metrics are persisted for chapter master and final encoded file.
FR-AUD-005	Define versioned export profiles instead of hardcoding one retailer's rules.	MUST	R1	Changing a profile creates a new version and does not alter prior validation results.
FR-AUD-006	Generate chapter files, opening/closing credits, full archive master, and optional M4B package with metadata/markers.	MUST	R1	Selected deliverables are reproducible from the approved revision.
FR-AUD-007	Produce a validation report with pass/fail, measured values, thresholds, warnings, profile version, checksums, and tool versions.	MUST	R1	The report is included in the export package and can be rerun independently.
FR-AUD-008	Retain raw/imported audio, provider outputs, selected takes, edited chapter masters, final masters, and exports according to retention policy.	MUST	R1	Asset inventory shows lineage and retention state for each file.

3.7 Quality control, issues, and approval

ID	Requirement	Priority	Release	Acceptance test
FR-QC-001	Transcribe selected chapter audio and compare it with the approved production script at word/sentence level.	MUST	R1	The QC run identifies insertions, deletions, substitutions, and low-confidence regions with time ranges.

ID	Requirement	Priority	Release	Acceptance test
FR-QC-002	Validate project pronunciation entries and flag inconsistent or suspect occurrences.	MUST	R1	The same approved term spoken differently across chapters creates a reviewable issue.
FR-QC-003	Run technical checks for clipping, true peak, loudness/RMS, noise floor, silence, distortion/artifact indicators, format, and duration anomalies.	MUST	R1	Threshold failures create typed issues tied to measured evidence.
FR-QC-004	Verify chapter/file completeness, order, credits, naming, and no missing/duplicated production segments.	MUST	R1	The exported file map reconciles to approved chapters and required credits.
FR-QC-005	Create issues with category, severity, confidence, time range, text range, owner, status, evidence, and resolution.	MUST	R1	A reviewer can filter, assign, resolve, reopen, and audit every issue.
FR-QC-006	Support blocking, critical, major, minor, and informational severities with configurable approval rules.	MUST	R1	Blocking issues prevent chapter/master approval.
FR-QC-007	Route low-confidence AI/ASR findings to human review; AI output remains a recommendation with model/prompt/version provenance.	MUST	R1	No low-confidence recommendation silently changes script, audio, or approval state.
FR-QC-008	Require named reviewer and producer approval on the frozen revision before export.	MUST	R1	Approval records the exact manuscript/script, EDL, assets, QC runs, and export profile versions.
FR-QC-009	Provide chapter and project QC dashboards showing unresolved issues, trends, false positives, and recheck status.	SHOULD	R2	Users can identify remaining work and recurring defect types.

3.8 Packaging, distribution, and post-release records

ID	Requirement	Priority	Release	Acceptance test
FR-PUB-001	Maintain audiobook metadata: title, subtitle, series, author, narrator credits/bio, description, keywords, categories, language, ISBN/identifiers, and runtime.	MUST	R1	Required fields are validated for each selected channel.
FR-PUB-002	Validate square cover presence, dimensions/format rules, and version association through a channel profile.	MUST	R1	Invalid or missing cover blocks package approval.
FR-PUB-003	Include audio files, metadata, cover, rights confirmation, disclosure, checksums, and validation report in a versioned export package.	MUST	R1	Export manifest lists every file and checksum and references the frozen approval.
FR-PUB-004	Generate exports only from a frozen, approved project revision; later edits create a new export candidate.	MUST	R1	Changing script/audio after approval cannot mutate an existing package.
FR-PUB-005	Store submission target, date, submitter, retailer response, errors, status, and published identifier even when submission is manual.	MUST	R1	A manually submitted package can be tracked through accepted/rejected/published states.
FR-PUB-006	Provide retailer/distributor adapters behind a DistributionProvider contract where reliable APIs or managed routes exist.	SHOULD	R2	A new adapter can submit a frozen package and poll/map external status without changing core workflow.

ID	Requirement	Priority	Release	Acceptance test
FR-PUB-007	Maintain channel eligibility rules for synthetic/hybrid narration and approved production routes.	MUST	R1	The system warns or blocks a channel when the project's narration/provider route is ineligible.
FR-PUB-008	Track post-release units, revenue, listen time, ratings, reviews, complaints, and rights/royalty obligations when data is available.	COULD	R2	Imported channel data is tied to the project and report period.

3.9 Analytics, audit, and production timeline

ID	Requirement	Priority	Release	Acceptance test
FR-ANA-001	Show project progress by stage, chapter, job, issue severity, approval, and export status.	MUST	R1	Dashboard values reconcile to underlying records.
FR-ANA-002	Track estimated and actual provider cost, characters/seconds generated, reused segments, retries, and cost per finished hour.	MUST	R1	Cost report reconciles to synthesis attempts and provider billing units.
FR-ANA-003	Maintain an append-only audit/event record for user commands, automated decisions, state changes, provider calls, approvals, exports, and distribution status.	MUST	R1	Every material change has actor/system, timestamp, entity, correlation ID, and payload.
FR-ANA-004	Provide a chronological production timeline that links events to jobs, assets, issues, costs, and approvals.	SHOULD	R2	A producer can replay a project from manuscript upload to published status.
FR-ANA-005	Export operational reports for pilot evaluation: elapsed time, human hours, defects, regeneration rate, cost, and channel result.	MUST	R0	The pilot produces a comparable report for every title.
FR-ANA-006	Build business analytics from the same domain events rather than a separate manual source of truth.	SHOULD	R2	Production and business reports trace back to event/entity IDs.

3.10 Administration and platform controls

ID	Requirement	Priority	Release	Acceptance test
FR-ADM-001	Manage users, invitations, roles, organization defaults, and project access.	MUST	R1	Revoked access is effective immediately and logged.
FR-ADM-002	Configure provider credentials, regions/endpoints, quotas, enabled capabilities, health, and cost tables with secrets stored outside application code.	MUST	R1	Secrets are never returned to clients or stored in plaintext application data.
FR-ADM-003	Configure upload, storage, retention, concurrency, budget, and rate limits.	MUST	R1	Limits are enforced by API/workers and produce clear errors/approval requests.
FR-ADM-004	Manage versioned export profiles, disclosure templates, pronunciation defaults, and naming conventions.	MUST	R1	Changes apply only to new runs/exports unless explicitly migrated.
FR-ADM-005	Define organization/project workflows and declarative rules for voice, provider, budget, QC severity, notification, and approval.	SHOULD	R2	A producer changes a pipeline or rule without a code deployment.

ID	Requirement	Priority	Release	Acceptance test
FR-ADM-006	Enable/disable provider, QC, analytics, and distribution plugins with typed configuration and health checks.	COULD	R3	A compatible plugin is installed for one tenant without redeploying core services.
FR-ADM-007	Support tenant isolation, quotas, usage billing, branding, and marketplace charges.	COULD	R3	Cross-tenant access tests fail at data, object, queue, and secret boundaries.

4. User Experience Specification

The existing React interface should evolve from upload/progress/download controls into a complete project workspace. The core interaction model is text-first: the manuscript and production script remain primary, while waveform, timing, voice settings, costs, jobs, and QC stay attached to the selected text.

Audiobook Project Workspace			
Project navigator chapters · speakers · assets	Text-aligned production editor script · sentence selection · waveform · playback cursor		Inspector voice · direction · pronunciation · cost
1. Opening credits 2. Chapter 1 3. Chapter 2 4. Chapter 3 Filters Needs review Failed jobs Blocking QC	Selected sentence "The corridor went silent before the door opened." [Regenerate selection] [Compare takes] [Keep take] Context is preserved from adjacent sentences; edits never overwrite the source manuscript.		Voice assignment Narrator: Premium US English Emotion: restrained tension Rate: 0.96x Pronunciation: corridor / ... /
	Waveform + transcript timeline word timestamps · issue markers · edit boundaries · playback		QC findings 1 possible word mismatch True peak: pass Silence: pass
	Chapter actions Preview Synthesize Run QC Approve		Job status Completed · 18 segments reused

Figure 2. Proposed R1 project workspace. The editor, audio timeline, voice inspector, QC queue, and job status share one context.

4.1 Required screens

Screen	Primary content and actions
Portfolio dashboard	Projects, stage, owner, target release, progress, cost, blocking issues, approval/export status.
New project wizard	Commercial metadata, target channels, narration mode, rights/consent, budget, provider posture, export profile.
Manuscript review	Upload status, parsed chapters, source/script diff, visual-reference queue, speaker/segment structure, completeness.
Voice & pronunciation	Voice search/filter, previews, assignments, direction controls, pronunciation dictionary, voice bible, rights status.
Production editor	Text, waveform, word timings, take comparison, punch-in regeneration, EDL, keyboard review, chapter actions.
Job monitor	Queued/running/completed/failed/cancelled jobs, attempts, provider, cost, progress, retry/cancel/resume.
QC console	Issue queue, filters, time/text evidence, severity, owner, resolution, recheck, chapter/project approval.

Screen	Primary content and actions
Mastering & export	Profile selection, measured metrics, warnings, validation report, deliverables, frozen revision, package manifest.
Distribution & status	Eligibility, disclosure, package, manual/API submission record, external status, errors, published identifiers.
Administration	Users, providers, secrets, quotas, budgets, profiles, retention, templates, audit, workflows/rules (R2).

4.2 Interaction requirements

- Progressive disclosure: advanced SSML, chunking, rate/pitch, mastering, and provider details remain available without overwhelming a first-time producer.
- Preview before commitment: any expensive or long-running operation presents scope, provider, estimated cost, reusable work, and expected output.
- Actionable errors: distinguish invalid input, provider rejection, quota, credential, policy, rights, QC, and processing failures.
- Accessible upload: drag/drop plus keyboard-accessible file picker; visible supported types/limits and upload/parse progress.
- Non-interruptive notifications: inline/toast success and error messages instead of browser alert dialogs.
- Responsive but desktop-first: editing and QC prioritize wide screens; dashboard and approvals remain usable on tablet.
- Keyboard review: play/pause, seek, loop, next issue, accept, reject, regenerate, and save mappings are documented and discoverable.
- Every AI recommendation displays confidence, provenance, and review state; accepting it is a user action.

4.3 Critical user journey: sentence correction

1. Reviewer clicks an ASR mismatch or selects a sentence in the production script.
2. Workspace loops the sentence with configurable context and shows the current take, speaker/voice, pronunciation, and direction.
3. User corrects script or settings, sees the estimated incremental cost, and requests regeneration.
4. The system creates a new segment job with an idempotency key and preserves adjacent context metadata.
5. When complete, the new take appears beside prior takes; the user A/B plays each take in chapter context.
6. Selecting a take creates a new edit decision and continuity check; it does not delete previous audio.
7. The affected QC checks rerun; the issue is resolved only when the reviewer confirms the evidence.

5. Technical Architecture

The proposed implementation preserves the existing React/TypeScript and Python/Flask investment, but replaces in-process threads and transient dictionaries with durable domain state, a real job queue, object storage, versioned APIs, and provider contracts. A framework rewrite is not required to make the product work.

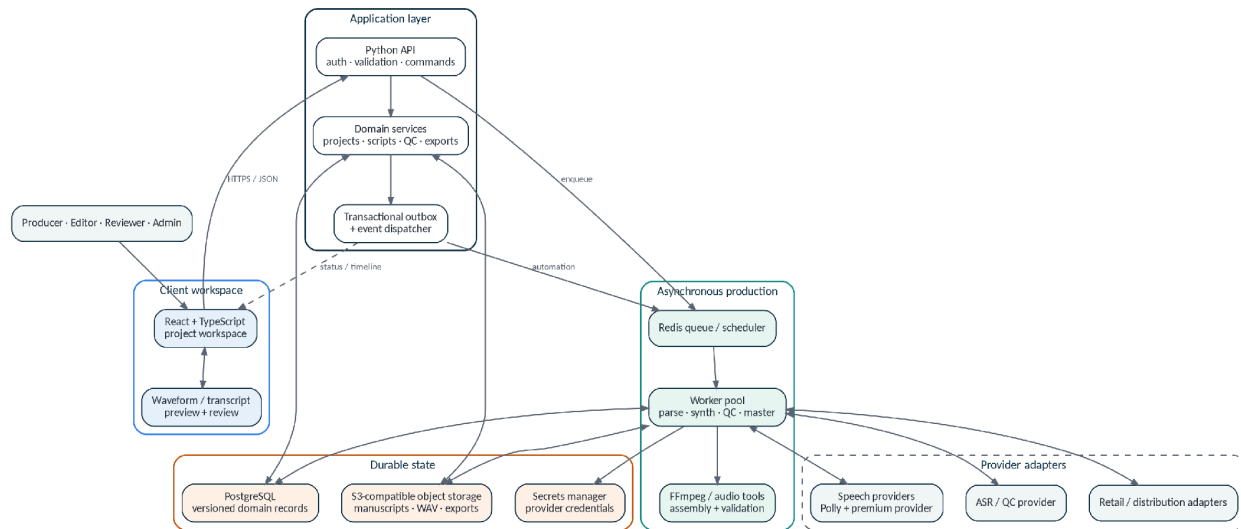


Figure 3. Proposed R1/R2 component architecture. Long-running work is isolated from the web process; all durable state and assets remain addressable and auditable.

5.1 Proposed implementation stack

Layer	Baseline	Release
Web client	React + TypeScript + Vite; reuse Tailwind/component structure; waveform/transcript component; generated API types.	R1
API	Existing Flask application refactored into route/application/domain/repository layers; OpenAPI 3 contract; Gunicorn production server.	R1
Database	PostgreSQL for projects, revisions, chapters, segments, jobs, issues, approvals, profiles, events, and cost records; JSONB only for bounded provider payloads.	R1
Queue / workers	Redis + Celery baseline (RQ is acceptable if team standard); separate queues for ingest, synthesis, QC, mastering, and distribution.	R1
Object storage	S3-compatible storage with organization/project prefixes, checksums, encryption, lifecycle policies, and presigned access.	R1
Audio processing	FFmpeg/ffprobe plus tested metering/validation wrappers; all processing invoked from workers with pinned versions.	R1
Speech / ASR	Provider adapters; Polly baseline, one premium provider, and an ASR/QC provider selected through RO.	R0/R1
Events	Transactional outbox table + dispatcher; idempotent consumers; event bus may begin as Redis/Celery and evolve later.	R1
Status updates	Polling every 3–5 seconds is acceptable for R1; server-sent events or WebSocket push is an R2 enhancement.	R1/R2
Deployment	Containerized services; development via Docker Compose; production on managed container/VM platform with managed Postgres, Redis, and object storage.	R1
Observability	Structured JSON logs, correlation IDs, job/provider metrics, error tracking, audit events, dashboards, alerts.	R1

Proposed implementation decision

Keep Flask for R1 to avoid a low-value rewrite. Publish an OpenAPI contract and enforce service boundaries. A later migration to FastAPI is optional and should be justified by measured team or performance needs, not aesthetics.

5.2 Service boundaries

Boundary	Owns
Identity & policy	Authentication, role checks, organization/project scope, rights/budget/approval policies.
Project service	Project metadata, lifecycle, targets, rights, disclosure, approvals, templates.
Content service	Upload, parse, revisions, chapters, segments, speakers, pronunciation, completeness.
Voice service	Provider catalog, previews, assignments, voice bible, capabilities, cost tables.
Production service	Job planning, content-addressed cache, synthesis attempts, provider routing, assets.
Edit service	Takes, edit decisions, EDL, assembly plan, continuity checks, chapter lock.
QC service	ASR diff, technical metrics, issue lifecycle, reviewer assignment, rechecks.
Mastering/export service	DSP pipeline, profiles, validation, package manifest, frozen export.
Distribution service	Eligibility, submission records, adapters, status mapping.
Event/audit service	Outbox, event dispatch, timeline projection, webhook/analytics consumers.

5.3 Asynchronous job model

State	Meaning
queued	Validated command persisted; waiting for an eligible worker.
planning	Scope, segments, cache hits, provider plan, estimate, and quotas resolved.
running	Worker is processing local or provider work.
awaiting_provider	External async task submitted; worker/poller awaits completion.
post_processing	Audio normalization, timings, checksums, metrics, or assembly.
completed	All required outputs and events persisted.
failed	Terminal attempt failure with typed error and retry eligibility.
cancellation_requested	User/system requested stop; worker must reach a safe checkpoint.
cancelled	No further work will be committed; partial external cost/provenance remains recorded.

- Every command accepts an idempotency key. The unique key scope includes organization, command type, target entity, and request hash.
- Workers update heartbeat/progress at safe checkpoints. A watchdog marks stale work and schedules recovery based on the job type.
- Provider attempts are immutable. A retry creates a new attempt and may select a different provider; it never overwrites the failed attempt.

- Cancellation cannot always cancel an already-billed external task. The platform stops polling/committing when possible and records any unavoidable cost.

5.4 Object and file lifecycle

Prefix	Contents / retention
source/	Original manuscripts and imported rights/consent evidence; immutable, long retention.
script/	Versioned normalized text, structure, dictionaries, direction, and manifests.
provider/	Raw provider outputs and timing payloads; immutable per attempt.
takes/	Normalized PCM/WAV segment takes and imported human pickups.
chapters/	Rendered chapter revisions and QC reference audio.
masters/	Approved archive master and profile-specific mastered assets.
exports/	Frozen package versions, encoded files, metadata, cover, manifest, validation report.
temp/	Ephemeral working files with short lifecycle and guaranteed cleanup.

object_key.txt

org/{organization_id}/project/{project_id}/{prefix}/{entity_id}/{version}/{filename}

Every stored object records: checksum, media type, byte length, creator/job, encryption, retention class, and lineage.

5.5 Reliability patterns

- Transactional outbox: domain change and event are committed in one database transaction; dispatch is retried independently.
- At-least-once delivery with idempotent consumers: duplicate events do not duplicate jobs, webhooks, costs, or analytics.
- Content-addressed reuse: approved audio is reused only when every synthesis-affecting input matches the cache key.
- Deterministic export: package manifests reference exact revisions, EDL, assets, profile version, tool versions, and checksums.
- Provider circuit breakers and health state: temporarily remove failing providers from new selection while preserving manual override and evidence.
- Resource cleanup: temporary local/S3 objects, completed task records, and generated presigned links follow explicit lifecycle policies.

6. Domain & Data Model

The central domain object is not an MP3. It is the versioned production record that links manuscript text to speakers, jobs, audio assets, edits, QC, approvals, exports, rights, cost, and events.

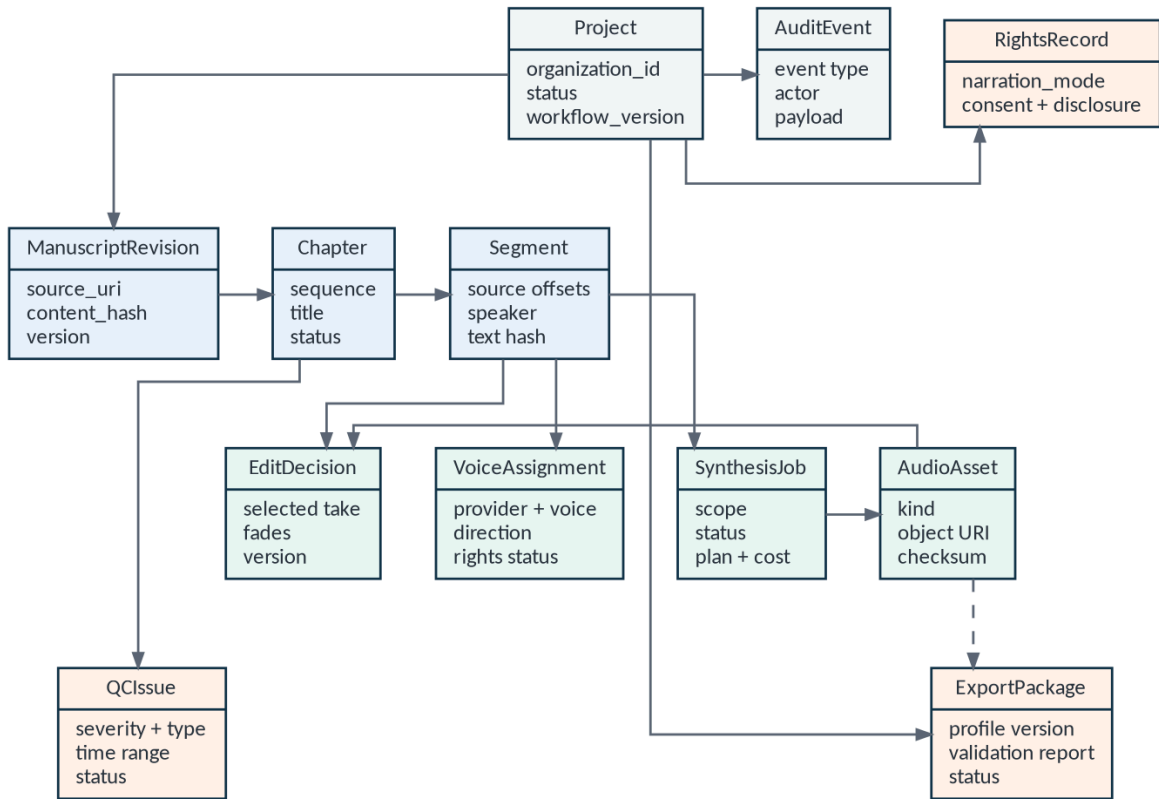


Figure 4. Simplified core domain. Additional tables support users, providers, rules, workflows, distribution, and analytics.

Entity	Responsibility
Organization	Security/configuration boundary in R1; contains users, projects, providers, profiles, and budgets.
Project	Title-level production container; status, targets, narration mode, workflow version, owner, budget.
RightsRecord	Manuscript/audio rights, consent, allowed uses, territory, term, compensation, credit, disclosure, approvals.
ManuscriptRevision	Immutable source or editable production-script revision with checksum and lineage.
Chapter / Segment	Ordered structure and smallest production unit; source offsets, text hash, type, speaker, direction.
Speaker / VoiceAssignment	Narrator/character identity and provider-independent voice mapping with rights status.
PronunciationEntry	Approved term, phonetic value, provider representation, scope, evidence, version.
SynthesisJob / Attempt	Requested scope, plan, status, idempotency, provider request/response, cost, timings, errors.
AudioAsset	Immutable file/object record for raw output, take, human pickup, chapter, master, or export.
EditDecision / ChapterRevision	Selected takes, ordering, trims, gaps, fades, continuity results, locked revision.
QCRun / QCIssue	Automated/human evaluation, metrics, findings, severity, time/text range, resolution.
Approval	Actor, role, entity/revision, decision, comments, timestamp, policy version.
ExportProfile / ExportPackage	Versioned thresholds and output rules; frozen deliverables, manifest, validation, status.

Entity	Responsibility
DistributionSubmission	Target channel, package, eligibility, submission method, external ID, response/status.
AuditEvent / OutboxEvent	Append-only domain event and reliable publication record with correlation/causation.

6.1 Versioning and immutability rules

- Original manuscripts, provider attempts, audio assets, QC results, approvals, and exports are immutable records.
- Corrections create new manuscript/script, chapter, edit, or profile versions and retain lineage to prior versions.
- Only one chapter revision may be marked current for a project stage; a frozen export references explicit revision IDs rather than “latest.”
- Deleting a project is a governed retention operation. Normal users archive; administrators execute policy-driven purge after retention/legal checks.
- Provider payloads may be retained as sanitized JSONB, but secrets and temporary signed URLs are never persisted as domain data.

7. API & Event Contracts

The frontend, workers, integrations, and later SDKs should share one versioned contract. R1 exposes REST resources and command endpoints under /api/v1. GraphQL is not required for the working release.

Method	Resource	Purpose
POST	/projects	Create project and baseline rights/target configuration.
GET/PATCH	/projects/{id}	Read or update allowed project fields.
POST	/projects/{id}/manuscripts	Upload and start parse job.
GET	/manuscripts/{id}/structure	Read chapters, segments, warnings, and source mapping.
POST	/manuscripts/{id}/revisions	Create corrected production-script revision.
PATCH	/chapters/{id}	Edit title/order/include/status or lock revision.
GET/POST	/projects/{id}/pronunciations	List/create dictionary entries.
GET	/providers/voices	List eligible voices and capabilities.
POST	/projects/{id}/voice-previews	Generate short preview.
PUT	/speakers/{id}/voice-assignment	Assign provider-independent voice profile.
POST	/synthesis-jobs	Start segment/chapter/project synthesis after estimate/approval.
GET	/jobs/{id}	Read state, progress, attempts, costs, outputs, and error.
POST	/jobs/{id}/cancel	Request cancellation.
POST	/jobs/{id}/retry	Retry eligible failed work.
POST	/segments/{id}/regenerations	Generate a new take for selected text/settings/context.
POST	/segments/{id}/take-selection	Commit a selected take as an edit decision.
POST	/chapters/{id}/qc-runs	Run selected automated QC suite.

Method	Resource	Purpose
GET/PATCH	/qc-issues/{id}	Read/assign/resolve/reopen issue.
POST	/chapters/{id}/approvals	Approve/reject frozen chapter revision.
POST	/exports	Create/validate frozen export package.
GET	/exports/{id}/validation	Read metrics, threshold results, and manifest.
POST	/distribution-submissions	Record or execute channel submission.
GET	/projects/{id}/events	Read audit/production timeline events.

7.1 API conventions

- Authentication and organization/project scope are enforced on every endpoint; identifiers never grant access by themselves.
- Commands that can cost money or create side effects require Idempotency-Key and return a resource/job ID.
- Errors use stable type codes, human guidance, correlation ID, retryability, and field-level details where applicable.
- Pagination, filtering, sorting, and sparse expansion are consistent across collections.
- The OpenAPI contract generates TypeScript client types and test fixtures; backend and frontend do not maintain duplicate hand-written models.

7.2 Domain event catalog

Event	Meaning
ProjectCreated	New production record initialized.
RightsStatusChanged	Rights/consent/disclosure state changed.
ManuscriptUploaded	Immutable source stored.
ManuscriptParsed	Structure and warnings produced.
ScriptRevisionApproved	Production script frozen for next stage.
VoiceAssigned	Speaker mapped to an approved voice profile.
SynthesisRequested	Billable production command accepted.
SegmentGenerated	A provider attempt produced a normalized take.
SynthesisFailed	Attempt failed with typed cause.
TakeSelected	Edit decision changed the current chapter assembly.
QCRunCompleted	QC metrics and findings persisted.
QCIssueOpened / Resolved	Issue lifecycle changed.
ChapterApproved	Named reviewer approved exact chapter revision.
ExportCompleted	Frozen package and validation report created.
DistributionSubmitted	Package submitted or manual submission recorded.
DistributionStatusChanged	External retailer/distributor status mapped.

8. Provider & Extension Contracts

Provider independence is a day-one architecture requirement. The first two providers may be implemented inside the main repository, but they must conform to the same contract later used by plugins.

speech_provider.py

```
class SpeechProvider(Protocol):
    name: str
    version: str
    capabilities: set[str]

    async def health(self) -> ProviderHealth: ...
    async def list_voices(self, language: str | None) -> list[VoiceDescriptor]: ...
    async def estimate(self, request: SynthesisRequest) -> CostEstimate: ...
    async def preview(self, request: PreviewRequest) -> AudioResult: ...
    async def synthesize(self, request: SynthesisRequest) -> SubmissionResult: ...
    async def poll(self, external_task_id: str) -> ProviderTaskResult: ...

    # Request includes normalized text/SSML, voice, output target, lexicon,
    # direction, context_before/context_after, and idempotency key.
    # Result includes provider/model/version, billed units, timings, raw asset,
    # warnings, and error mapping.
```

8.1 Capability negotiation

Capability	Meaning
long_form	Provider supports chapter-scale or async synthesis.
ssml	Provider accepts supported SSML subset.
word_timestamps	Provider returns word/speech marks.
style_prompt	Provider accepts natural-language direction.
emotion_controls	Provider supports explicit emotion/style parameters.
custom_lexicon	Provider accepts reusable pronunciation lexicons.
voice_clone	Provider offers approved custom/replica voice capability.
streaming	Provider can stream preview/audio.
async_task	Provider returns external task ID and requires polling/callback.

8.2 AI agent contract

ai_agent.py

```
class AIAgent(Protocol):
    agent_key: str
    version: str

    async def analyze(self, input_ref: EntityRef, context: AgentContext) -> AIInference: ...
```

```
# AIInference stores model, model_version, prompt_version, input_hash,
# structured output, confidence, review_status, project_id, and timestamps.
# Output is always a recommendation; it never overwrites source content.
```

8.3 Distribution provider contract

distribution_provider.py

```
class DistributionProvider(Protocol):
    channel: str
    capabilities: set[str]

    async def validate_eligibility(self, project, package) -> EligibilityResult: ...
    async def submit(self, package, metadata, disclosure) -> SubmissionResult: ...
    async def status(self, external_id: str) -> DistributionStatus: ...

# Manual channels implement validation + status recording even when no API exists.
```

8.4 Plugin lifecycle (R3)

- registered → installed → enabled → disabled → uninstalled
- Typed, tenant-scoped configuration; secrets stored in the secrets manager.
- Contract tests, security scan, semantic-version compatibility, sandbox trial, health monitoring, and revocation.
- Plugins advertise capabilities so workflows, rules, and selectors can determine eligibility without hardcoded branches.

9. Audio Pipeline & Export Profiles

The mastering system must be reproducible, profile-driven, and evidence-producing. Generated MP3 chunks are not the internal master. Provider output is normalized to a lossless working format, assembled non-destructively, mastered once, measured, and encoded to each export target.

Step	Behavior
1. Normalize input	Decode provider/imported audio; validate duration/channels; convert to working PCM/WAV; preserve raw asset.
2. Segment continuity	Trim, align, apply intended gaps/crossfades, and run replacement-boundary checks.
3. Chapter assembly	Render EDL in sequence with credits, room tone, chapter gaps, and deterministic markers.
4. Cleanup	Optional de-click/de-plosive/de-noise/de-ess/EQ/dynamics stages with conservative defaults and audit.
5. Loudness & peak	Apply spoken-word loudness/RMS target and true-peak limiter without clipping or excessive pumping.
6. Validate master	Measure technical metrics and compare to versioned profile; fail or warn by rule.
7. Encode deliverable	Encode once from approved master to MP3/M4B/WAV profile; embed metadata/markers.
8. Package evidence	Write manifest, checksums, profile/tool versions, measured values, warnings, and approvals.

9.1 ACX research profile

Measurement	ACX NATION research baseline	Implementation rule
RMS loudness	-22 to -18 dB; research target approximately -20 dB.	Implement as configurable min/max/target and measure on final encoded chapter file.
True peak	No higher than -3 dB in the research summary. The question bank also cites -3.5 dB as a starting point.	Release gate: verify official current requirement; retain profile version and conservative headroom.
Noise floor	Research summary prints “≤60 dB RMS”; the question bank prints “≤ -60 dB.”	Release gate: treat the sign/metric discrepancy as unresolved until checked against official current documentation.
Sample rate	44.1 kHz.	Resample master/deliverable through pinned, tested settings.
Channels	Mono or stereo.	Default to mono narration unless project/profile requires stereo.
Codec / bitrate	MP3 192 kbps CBR.	Encode once from lossless approved master; validate actual codec/bitrate.
Head / tail room tone	1–5 seconds.	Profile controls exact minimum/maximum and credit/chapter behavior.

Release gate: current retailer rules

The policy and technical figures in ACX NATION are research inputs current to mid-2026, not a substitute for official submission documentation. The product must store versioned channel profiles and require an explicit “verified on” date/source before a production profile can be marked active.

9.2 Validation report fields

- Project, chapter/file, export profile ID/version, source approval ID, EDL/revision IDs, generated timestamp.
- Duration, sample rate, channels, codec, bitrate mode/value, RMS/loudness, sample peak, true peak, noise floor, clipping, silence/room tone, file size, checksum.
- Threshold, measured value, tolerance, pass/fail/warn, calculation method/tool/version, and diagnostic evidence.
- Completeness: expected vs actual chapter/credits/files, filename order, metadata, cover, rights, consent, and disclosure status.
- Approval and override record; overrides require authorized role, reason, and audit event.

10. Security, Rights & Governance

The source research makes platform eligibility, narrator consent, and disclosure product requirements rather than legal footnotes. The backend must prevent unauthorized spend, voice use, publication, and cross-project data access.

Control	Requirement
Authentication	Session/OIDC-based user authentication; secure cookies/tokens; no anonymous synthesis in production.
Authorization	Organization/project role policy on every query/command and object URL; deny by default.
Secrets	Provider credentials stored in managed secrets, scoped by environment/organization, rotated, never exposed to browser.

Control	Requirement
Data protection	TLS in transit; encrypted database/object storage; least-privilege IAM; private buckets; expiring presigned URLs.
Upload safety	MIME/extension/size checks, secure filenames, parser sandboxing/limits, malware scanning when public uploads are enabled.
Cost control	Budgets, quotas, rate limiting, concurrency limits, estimate-before-commit, alert thresholds, per-provider caps.
Voice rights	Consent and permitted-use evidence required for custom/replica voices; audit access and revocation workflow.
Disclosure	Versioned channel text/metadata and publishing gate; never inferred silently from provider name.
Audit	Append-only events for rights, provider use, AI inference, approvals, exports, and distribution.
Retention	Policy by asset class; legal hold; temporary cleanup; documented purge; no indefinite S3/local accumulation.
Tenant readiness	All R1 records and object paths carry organization scope; R3 adds tenant as billing/isolation boundary.

10.1 Voice rights record

- Identity of narrator/rights holder and verification status.
- Voice source and method: live performance, AI-assisted pickup, replica/clone, provider stock voice, or custom model.
- Permitted titles/series, media, territories, languages, channels, term, exclusivity, sublicensing, and derivative use.
- Compensation: fee, per-finished-hour, royalty share, minimums, reporting, and payment status.
- Required credit and disclosure language.
- Revocation/termination, model/data deletion obligations, posthumous/estate restrictions, and dispute contact.
- Signed agreement/evidence checksum and approval history.

11. Non-Functional Requirements

The values below are proposed engineering targets for the pilot and working release. They should be validated with real titles and updated through change control.

ID	Quality attribute	Acceptance target
NFR-001	Durability	Committed project state and object references survive API/worker restarts; database backups and object versioning are tested.
NFR-002	Job recovery	No accepted job disappears. Stale work is detected; retry/resume is safe and idempotent.
NFR-003	Interactive performance	Normal project/read APIs target ≤500 ms P95 excluding file transfer/provider calls; local UI interactions feel immediate.
NFR-004	Status freshness	R1 polling exposes job state changes within 5 seconds under normal load.
NFR-005	Preview target	Short preview request starts playback or returns a clear provider status within a pilot-validated target; proposed initial target ≤20 seconds.
NFR-006	Upload scale	Support at least 50 MB uploads and a 100k-word / 50-chapter title without losing structure or blocking the web process.

ID	Quality attribute	Acceptance target
NFR-007	Concurrency	Worker concurrency and provider rate limits are configurable; one project cannot starve all other work.
NFR-008	Availability	Internal R1 target 99.0% monthly API availability; production R2 target 99.5%, excluding provider outages and planned maintenance.
NFR-009	Security	No high/critical unresolved vulnerability at release; secrets scanning, dependency scanning, access tests, and audit tests pass.
NFR-010	Accessibility	Core web flows meet WCAG 2.1 AA intent: keyboard operation, focus, labels, contrast, error association, and captions/text alternatives.
NFR-011	Observability	Every request/job/provider call has correlation ID; structured logs, metrics, traces or timing spans, and alertable failure rates.
NFR-012	Cost safety	All billable commands are attributable, estimable, quota-controlled, and visible by project/provider.
NFR-013	Portability	Development runs in containers; provider interfaces and S3-compatible storage avoid unnecessary cloud lock-in.
NFR-014	Data privacy	Uploaded manuscripts and voice data are not used for model training unless an explicit agreement and project setting allow it.

12. Test & Acceptance Strategy

A tool that “generates audio” is not done. Release acceptance must prove end-to-end completeness, editability, quality control, rights enforcement, technical validation, recovery, and reproducibility on representative long-form titles.

Test level	Coverage
Unit	Parsers, segmenter, normalization, cost math, cache keys, policies, state transitions, metric parsers, filename rules.
Contract	SpeechProvider, ASR/QC, DistributionProvider, object storage, webhook/event schemas; run against mocks and provider sandboxes.
Integration	API + database + queue + worker + storage + FFmpeg; restart/retry/cancel scenarios; outbox delivery.
End-to-end	Browser journey from project creation and upload through sentence correction, QC, approval, export, and re-open.
Golden audio	Known input/takes with expected duration, checksums where deterministic, metric ranges, joins, markers, and validation result.
Long-form	90k–100k word / 35–50 chapter book; provider limits, order, completeness, memory, queue recovery, and export.
QC evaluation	Curated misreads, missing words, pronunciations, clipping, silence, peak, loudness, and artifact samples; measure false positives/negatives.
Security	Authorization matrix, IDOR/cross-project access, secret exposure, upload abuse, rate limit, cost abuse, presigned URL expiry.
Accessibility	Keyboard-only production flow, focus order, screen-reader labels, color contrast, error messaging, reduced-motion behavior.

Test level	Coverage
User acceptance	Producer/editor/proofer complete pilot titles and sign off on workflow, time savings, correction quality, and evidence package.

12.1 Canonical R1 acceptance scenario

1. Create a synthetic or hybrid audiobook project targeting at least ACX/Audible plus one other channel; complete rights and disclosure fields.
2. Upload a 90k–100k word DOCX with approximately 40 chapters, dialogue, unusual names, front/closing credits, and at least one visual-only reference.
3. Correct chapter detection, build speakers and pronunciation entries, approve the production script, assign voices, and generate previews.
4. Run full synthesis, terminate one worker, trigger one provider error, cancel/retry one chapter, and verify no duplicate billable output or lost state.
5. Identify a scripted mismatch, regenerate only that sentence, A/B compare takes in context, select one, and pass continuity recheck.
6. Run ASR-diff and technical QC; resolve all blocking issues; obtain named reviewer and producer approval.
7. Create a frozen ACX research-profile export, validation report, metadata/cover/rights/disclosure bundle, and checksum manifest.
8. Reopen the project and prove the final output can be reconstructed from revisions, assets, EDL, profile, tools, and approvals.

12.2 Release blockers

- Missing/duplicated text or chapters; destructive edits; untraceable selected audio; unrecoverable job state.
- Ability to synthesize or export with incomplete rights/consent/disclosure or unresolved blocking QC.
- Provider credentials exposed to client/logs; unauthorized cross-project access; uncontrolled billable endpoints.
- Master/export metrics not reproducible, profile thresholds not versioned, or official target specs not reverfied.
- Pilot users cannot correct a sentence in context without regenerating a chapter or using external tools for the core workflow.

13. Delivery Roadmap

The roadmap merges the “pilot before build” recommendation with the Platform Blueprint waves. Each release must deliver standalone production value; platform/ecosystem work cannot displace the editor, QC, mastering, and rights foundation.

Release	Build focus	Exit outcome
R0 · Evidence pilot	Select premium provider; produce 3–5 backlist titles; verify policies/specs; measure cost, time, defects, trust/feedback; validate punch-in and ASR QC.	Decision report and provider/export baselines.
R1A · Durable core	Project/rights/auth, Postgres/object storage, manuscript revisions, parse/structure editor, jobs/queue, Polly + premium adapter, previews.	Persistent project through first chapter generation.
R1B · Differentiated workflow	Text-aligned editor, takes, punch-in, EDL, human pickup, timing, cost/reuse, job control.	Producer can correct and lock a chapter.
R1C · Quality and delivery	ASR/technical QC, issue workflow, approvals, mastering, versioned profiles, package/manifest, audit report.	Working publisher tool and canonical acceptance pass.

Release	Build focus	Exit outcome
R2 · Production platform	Configurable workflow/rules, provider selector/failover, distribution adapters, analytics, timeline, hardened SLOs/security.	Multi-process production across titles/providers/channels.
R3 · Audiobook OS	Plugins, certification, marketplace, public API/webhooks/SDKs, tenant isolation, quotas, billing, custom branding.	Commercial ecosystem platform.

13.1 Backlog order within R1

1. Durable project, rights, auth, object storage, database, queue, and audit foundation.
2. Manuscript upload/parsing, script revision, chapter correction, speakers, pronunciation, completeness.
3. Provider contract, Polly adapter, premium adapter, voice previews, assignments, cost estimation, persistent jobs.
4. Text/waveform editor, timings, sentence regeneration, take comparison, EDL, assembly, continuity.
5. ASR diff, technical metering, issue workflow, approvals, rechecks.
6. Mastering, versioned export profiles, validation report, package manifest, retailer eligibility/disclosure.
7. Operational hardening, security tests, long-form acceptance, pilot evaluation reporting.

Scope guardrail

The 100-knob voice instrument, proprietary voice-model training, soundscaping, marketplace, GraphQL, billing, and broad retailer automation do not enter R1 unless they are required to pass the canonical acceptance scenario.

14. Risks & Open Decisions

The research contains several intentionally unresolved or fast-moving areas. They should become tracked product decisions and release gates rather than hidden assumptions.

Risk	Impact	Mitigation / decision
Retailer policy volatility	AI narration acceptance, disclosure, and approved production routes can change.	Versioned eligibility/disclosure profiles; verified-on/source fields; pre-release legal/operations check.
Audio spec discrepancy	ACX NATION contains -3 vs -3.5 peak and signed/unsigned noise-floor figures.	Do not silently normalize; verify official current figures and activate only reviewed profile version.
Voice rights / consent	Replica/cloned voices can create contractual, union, estate, and revocation risk.	Rights record, evidence, backend gate, channel policy, audit, deletion/revocation workflow.
Provider quality drift	Model/version changes can alter pronunciation, timing, or long-form consistency.	Pin/version where possible; golden samples, provider/model provenance, regression suite, approval before upgrade.
Long-form prosodic drift	Segment/chapter generation can flatten or change identity across hours.	Context-aware chunking, voice bible, reference/style metadata, sample gates, continuity QC, provider comparison.
Punch-in seams	Regenerated sentence may not match tone, room, timing, or prosody.	Context conditioning, adjacent playback, loudness/spectral checks, crossfade, take comparison, human override.
ASR false positives	Automated text diff can burden proofers or miss nuanced errors.	Confidence thresholds, issue severity, curated evaluation set, human review, false-positive analytics.
Cost overrun / abuse	Public or faulty jobs can trigger uncontrolled provider charges.	Authentication, quotas, rate limits, estimate/approval, idempotency, provider caps, anomaly alerts.

Risk	Impact	Mitigation / decision
Listener trust	Quality may be acceptable while willingness to try AI narration remains lower.	Consent/credit-forward metadata, clear channel disclosure, backlist pilot, hybrid option, listener feedback metrics.
Scope expansion	Platform/marketplace/voice-R&D work can delay a working production workflow.	R1 acceptance scenario and scope guardrail; release funding tied to working title completion.

14.1 Decisions required before R1 implementation lock

Decision	Required outcome
Premium provider for R0/R1	Choose after sample quality, long-form continuity, timing support, rights terms, price, and channel eligibility tests.
Initial channel set	Confirm whether wide distribution through INaudio or other aggregator is required; it can constrain production route.
Narration posture	Decide default pilot mix: synthetic, hybrid, or narrator-replica; define acceptable title categories/backlist.
Official export profiles	Verify current ACX/Audible, Apple, Spotify, Google Play, and any distributor requirements and sources.
Data residency / deployment	Internal cloud only, managed SaaS, or self-host option; voice/manuscript privacy obligations.
Retention	How long to retain raw provider outputs, human voice samples, consent evidence, masters, and failed attempts.
OCR and PDF scope	Text-based PDF in R1; decide whether scanned-PDF OCR is required by actual catalog inputs.
Availability/SLO	Confirm internal-only vs customer-facing R1 and adjust availability/on-call targets.

15. Definition of Done

15.1 Working tool completion test (R1)

- A producer can complete the canonical 90k–100k word project from manuscript to frozen, validated export inside the application.
- The system preserves immutable source, editable script revisions, voice/rights decisions, job attempts, costs, takes, EDL, QC, approvals, and export evidence.
- A single sentence can be corrected and regenerated in context without rerendering the chapter, losing prior takes, or performing the core edit in an external DAW.
- Automated QC and human review identify, resolve, and recheck text/audio and technical issues; blocking findings prevent approval.
- A server/worker/provider failure can be recovered without losing accepted work or duplicating output silently.
- Every billable action is authenticated, scoped, estimated, idempotent, attributable, and subject to budget/quota controls.
- The final package is generated only from explicit frozen revisions and includes profile/version, metrics, manifest, checksums, rights, disclosure, and approvals.

15.2 Platform completion test (R3)

- A publisher can define and run its production workflow without a code change.
- Voice, budget, routing, QC, compliance, and notification policy are enforced by declarative rules.
- AI agents produce reviewable, versioned recommendations that never overwrite source content.
- A speech provider, QC service, AI capability, or retailer can be added through stable plugin contracts without redeploying core logic.
- Providers are selected by policy, cost, quality, latency, capability, and availability with automatic failover.
- Approved masters can be distributed and tracked across supported channels; production and business analytics reconcile to the event source.
- Customers can operate the platform through public API, webhooks, and SDKs.
- Multiple tenants are served with strict isolation, quotas, billing, and tenant-scoped plugins/configuration.

One-sentence platform test

Audiobook OS is complete when it can add a new speech provider, a new distribution channel, and a new AI capability for a customer with a different production process without redesigning the product or shipping bespoke core code.

Appendix A · Source-to-Spec Traceability

Source	Key contribution	Specification coverage
Audiobook OS Platform Blueprint §§1–3	Persistent production record; five pillars; configuration, contracts, events, governance, traceability.	Executive Summary; §§1–3, 5–7, 10–11, 15.
Blueprint §§4–6	Workflow engine, declarative rules, AI Service Mesh, provenance, human review.	§§2–3, 5.5, 7.2, 8.2, 10, 13.
Blueprint §§7–9	Plugin contract, marketplace model, provider selector/failover.	§§3.3–3.4, 8, 13, 15.2.
Blueprint §§10–17	Governance, distribution, analytics, optimization, timeline, events, API/SDK, multi-tenancy.	§§3.8–3.10, 5–7, 10–11, 13, 15.2.
Blueprint §§18–20	Delivery waves and platform completion definition.	Executive release ladder; §§13 and 15.
ACX NATION Tab 2 / pp. 4–8	Policy/disclosure, voice replica model, audience trust, build-vs-buy, technical spec, provider landscape, pilot recommendation.	Executive Summary; §§1, 3.1, 3.3, 3.8, 9, 10, 13–14.
ACX NATION Tab 4 / pp. 10–25	Eight-phase production system, script engineering, narrator acquisition, recording, engineering, QA, packaging, business assets.	§§1–4, 6, 9, 12.
ACX NATION Tab 3 / pp. 27–39	Research questions on long-form TTS, soundscaping, mastering, text-first editing, scaling, policy/rights, QC, competitors.	Scope guardrails; §§3–5, 9, 11–14.
ACX NATION Tab 5 / pp. 41–56	Polly setup, async long-form synthesis, SSML, chunking, S3, cost/caching, custom voice considerations.	§§3.3–3.4, 5, 8–9, 11.
ACX NATION Tab 6 / pp. 58 onward	Full-stack gaps: formats, file parsing, settings, SSML, preview, cancel, persistent jobs, UI integration, security, cleanup, performance.	§§3–5, 7, 10–12.
ACX NATION Tab 9	100-knob real-time voice synthesizer research platform and long implementation plan.	Explicitly deferred from R1; referenced in scope guardrail and risk control.

Appendix B · Glossary

Term	Definition
ASR	Automatic speech recognition; used to transcribe generated/recorded audio for text comparison.
EDL	Edit decision list; non-destructive instructions selecting assets, trims, ordering, gaps, and fades.
Provider adapter	Implementation translating the platform contract into a speech, ASR, QC, or distribution vendor API.
Production script	Editable narration-ready text derived from, but independent of, the immutable source manuscript.
Segment	Smallest addressable unit for voice assignment, generation, editing, timing, QC, and reuse.
Take	One version of audio for a segment, produced by a provider or human recording.
Punch-in regeneration	Regenerating only selected text and replacing it non-destructively in context.
Content-addressed reuse	Reusing an approved asset only when all synthesis-affecting inputs have the same hash.
Export profile	Versioned output rules and thresholds for a channel or archival format.
Frozen revision	Explicit, immutable set of script, assignments, EDL, assets, QC, and approvals used to create an export.
True peak	Peak estimate accounting for inter-sample peaks; distinct from sample peak.
PFH	Per finished hour; common audiobook production/compensation unit.
AI inference	Versioned recommendation with model/prompt/input provenance, confidence, and human review status.
Transactional outbox	Reliability pattern that stores a domain event in the same database transaction as the state change.

Appendix C · Approval Record

Approval role	Name	Decision	Date
Product owner			
Engineering lead			
Publishing / production lead			
Legal / rights reviewer			